Week 07: Wednesday, AST 5011: Astrophysical Systems

Compact Object Physics: White Dwarf Cooling & Neutron Star Measurements

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With contributions totally ripped off from Carl Fields (UA), Mike Zingale (SUNY), Cole Miller (UMD), and Abi Nolan (Purdue).

Part 1: White Dwarf Cooling

Why Do White Dwarfs Cool?

Recall from Lecture 4 that white dwarfs are supported by electron degeneracy pressure, with a mass-radius relation $R \sim M^{-1/3}$ and a maximum mass (the Chandrasekhar limit) of $\sim 1.4\,M_\odot$. From Lecture 7, we know their cores are predominantly carbon and oxygen.

Unlike main-sequence stars, white dwarfs have no nuclear energy source. Degeneracy pressure is independent of temperature, so there is no contraction either. The only energy reservoir is the thermal energy of the ions in the core. As this heat radiates away through the thin envelope, the white dwarf gradually fades.

The key question: how fast does a white dwarf cool? The answer turns out to constrain the age of the Galactic disk.

The Two-Zone Structure

A cooling white dwarf has a simple two-zone structure:

1. A degenerate core containing nearly all the mass. Electron conduction is so efficient that the core is nearly isothermal at temperature T_c.

2. A thin radiative envelope of non-degenerate ideal gas surrounding the core. This envelope is opaque and controls the luminosity — it acts as an insulating blanket.

The transition between core and envelope occurs where the Fermi energy equals the thermal energy: $E_F \sim kT$. Because the envelope is thin, L is approximately constant through it.

The luminosity of the white dwarf is set entirely by the envelope's ability to radiate, which depends on the core temperature T_c.

Envelope Structure and the L-T Relation

The radiative envelope obeys the usual temperature gradient:

$$\nabla_{\rm rad} = \frac{d\log T}{d\log P} = \frac{3}{16\pi acG}\,\frac{\bar{\kappa}\,P}{T^4}\,\frac{L_\star}{M_\star}$$

Using a bound-free opacity of the form $\kappa_{\rm bf} \sim \kappa_0\,\rho\,T^{-7/2}$ and an ideal gas equation of state $P = \rho kT/(\mu m_u)$, one can integrate from the surface inward. The result is a power-law relationship between pressure and temperature in the envelope, giving:

$$\frac{L_\star}{L_\odot} \approx 6.5\times10^{-3}\,\mu\left(\frac{M_\star}{M_\odot}\right)T_{c,7}^{7/2}$$

where $T_{c,7} = T_c/10^7\,\mathrm{K}$ is the core temperature in units of 10^7 K and μ is the mean molecular weight in the envelope.

The luminosity depends very steeply on the core temperature ($L \propto T_c^{7/2}$). A small drop in core temperature causes a large drop in luminosity.

The Mestel Cooling Law

The thermal energy of the white dwarf is stored in the ions (degenerate electrons carry negligible thermal energy because only a fraction $\sim kT/E_F$ can participate). The specific heat per unit mass is:

$$c_V = \frac{3}{2}\,\frac{k}{A\,m_u}$$

where A is the mean atomic weight of the ions (12 for carbon, 14 for a C/O mix).

Energy conservation gives:

$$L_\star = -\frac{dE_{\rm th}}{dt} = -c_V\,M_\star\,\frac{dT_c}{dt}$$

Substituting the $L \propto T_c^{7/2}$ envelope relation and integrating, we obtain the Mestel cooling law:

$$\boxed{t_{\rm cool} \approx 6\times10^6\ \mathrm{yr}\left(\frac{A}{12}\right)^{-1}\left(\frac{M}{M_\odot}\right)^{5/7}\left(\frac{\mu}{2}\right)^{-2/7}\left(\frac{L_\star}{L_\odot}\right)^{-5/7}}$$

The key scaling is $L \propto t^{-7/5}$. More massive WDs cool slower (larger heat reservoir). Fainter WDs are older.

Crystallization Effects on Cooling

From Lecture 4, when the Coulomb coupling parameter $\Gamma = (Ze)^2/(r_{\rm ion}\, kT) \approx 170$, the ions crystallize into a lattice. In WDs this happens when the core cools to roughly 5×10^6 K.

Crystallization affects WD cooling in two important ways:

1. Latent heat release: the liquid-to-solid phase transition releases energy $\sim kT$ per ion. This is comparable to the thermal energy already stored, providing an extra energy source that slows cooling by roughly a factor of 2 at the crystallization epoch.

2. Gravitational energy from composition separation: in a C/O white dwarf, the solid phase preferentially incorporates oxygen (higher charge, stronger Coulomb binding). The lighter carbon-enriched liquid rises, releasing gravitational potential energy. This provides yet another energy source, further delaying cooling by ~ 1–$2\,\rm Gyr$ for massive WDs.

These effects create a pile-up of white dwarfs at specific luminosities in the WD luminosity function — a prediction that has been spectacularly confirmed by *Gaia* observations.

The WD Luminosity Function: A Cosmic Clock

If we observe many white dwarfs in a stellar population and count how many exist at each luminosity, we construct the WD luminosity function $n(L)$.

From the Mestel law, $L \propto t^{-7/5}$, so:

$$\frac{dt}{dL} \propto L^{-12/7}$$

At low luminosities (old WDs), WDs spend more time per luminosity interval, so $n(L)$ rises toward faint magnitudes. But there must be a sharp cutoff at the luminosity corresponding to the age of the population — no WD can be fainter than one that has been cooling since the population formed.

Observations show this cutoff at $\log(L/L_\odot) \approx -4.5$, corresponding to a cooling age of ~ 8–$10\,\rm Gyr$. This provides an independent lower limit on the age of the Galactic disk — one of the most elegant applications of stellar physics to cosmology.

WD cosmochronometry is complementary to other age-dating methods (main-sequence turnoff in globular clusters, radioactive dating, CMB).

Exercise: White Dwarf Cooling Timescale

The Mestel cooling law gives the time for a white dwarf to cool to luminosity $L_\star$:

$$t_{\rm cool} \approx 6 \times 10^6 \ {\rm yr} \left(\frac{A}{12}\right)^{-1} \left(\frac{M}{M_\odot}\right)^{5/7} \left(\frac{\mu}{2}\right)^{-2/7} \left(\frac{L_\star}{L_\odot}\right)^{-5/7}$$

Tasks:

1. Compute $t_{\rm cool}$ for a $0.6\,M_\odot$ C/O white dwarf ($A = 12$, $\mu = 2$) at $L = 10^{-2}\,L_\odot$, $10^{-3}\,L_\odot$, and $10^{-4}\,L_\odot$.
2. Plot $L/L_\odot$ vs $t_{\rm cool}$ (in Gyr) for masses $M = 0.4, 0.6, 0.8, 1.0\,M_\odot$. Use a log scale for L.
3. At what luminosity does a $0.6\,M_\odot$ WD reach a cooling age equal to the age of the Galactic disk ($\sim 10\,{\rm Gyr}$)?
4. Estimate the crystallization temperature for a C/O WD with $\bar{\rho} \approx 10^6\ {\rm g\ cm^{-3}}$ using $\Gamma = 170$, and compute the luminosity at which crystallization begins using the L-T_c relation.

▶ Solution

White Dwarf Binary Systems

Many white dwarfs exist in binary systems. When two WDs orbit each other at short periods, they lose energy through gravitational wave emission. The orbit shrinks, the period decreases, and the system evolves toward either a merger or the onset of mass transfer.

These compact WD binaries are important for several reasons:

1. They are guaranteed sources for the future space-based gravitational wave observatory LISA, which will be sensitive to millihertz GW frequencies — exactly where compact WD binaries radiate.

2. If the lighter WD fills its Roche lobe, mass transfer begins. If the accretor is a C/O WD and accumulates enough helium or hydrogen, the result can be a Type Ia supernova, a nova, or a stable AM CVn system.

3. The orbital decay from GW emission has been directly measured in several systems, providing beautiful confirmations of general relativity beyond the Hulse-Taylor binary pulsar.

Gravitational Wave Emission

A circular binary of component masses m_1 and m_2 with orbital period P emits gravitational waves at frequency $f_{\rm GW} = 2/P$ (twice the orbital frequency). The dimensionless strain amplitude at distance d is:

$$h = \frac{4}{d} \left(\frac{G\,\mathcal{M}_c}{c^2}\right)^{5/3} \left(\frac{\pi\,f_{\rm GW}}{c}\right)^{2/3}$$

where the chirp mass is:

$$\mathcal{M}_c = \frac{(m_1\, m_2)^{3/5}}{(m_1 + m_2)^{1/5}}$$

The orbit decays as energy is carried away by gravitational radiation. The Peters (1964) formula for the period derivative is:

$$\dot{P} = -\frac{192\pi}{5}\left(\frac{2\pi\, G\, \mathcal{M}_c}{c^3\, P}\right)^{5/3}$$

For the eclipsing double WD ZTF J1539+5027 ($P = 6.91$ min, $m_1 = 0.61\, M_\odot$, $m_2 = 0.21\, M_\odot$, $d \approx 2.4$ kpc), the predicted $\dot{P}$ matches observations to within measurement uncertainty — a direct detection of orbital decay from gravitational wave emission.

Mass Transfer and AM CVn Systems

As the orbit shrinks, the lighter (larger) WD may eventually fill its Roche lobe and begin transferring mass to the heavier WD. The outcome depends on the mass ratio $q = m_{\mathrm{donor}}/m_{\mathrm{accretor}}$:

- If q is small enough, mass transfer is stable: the donor loses mass, expands (recall $R \propto M^{-1/3}$ for WDs), and the system reaches a stable configuration where the transfer rate is set by GW-driven orbital evolution. These are the AM CVn systems — ultra-compact binaries with hydrogen-deficient (helium-rich) accretion.

- If q is too large, mass transfer is dynamically unstable, potentially leading to a merger. The merger of two C/O WDs near the Chandrasekhar mass is one channel for Type Ia supernovae (the double-degenerate scenario).

Known LISA verification binaries include:

System	Type	P_{orb} (min)	f_{GW} (mHz)
HM Cancri	Interacting DWD	5.4	6.2
ZTF J1539+5027	Eclipsing DWD	6.9	4.8
SDSS J0651+2844	Eclipsing DWD	12.8	2.6
AM CVn	AM CVn prototype	17.1	2.0

LISA: Observing WD Binaries in Gravitational Waves

LISA (Laser Interferometer Space Antenna) is an ESA/NASA mission planned for the mid-2030s. Three spacecraft in a triangular formation with 2.5 million km arms will detect gravitational waves in the millihertz band (0.1–100 mHz) — exactly where compact WD binaries radiate.

LISA will detect tens of thousands of individual WD binaries in the Milky Way. The tens of millions of unresolved binaries create a foreground noise below ~ 2–3 mHz. Several dozen known systems are guaranteed detections (verification binaries) — their parameters are already known from electromagnetic observations.

For each resolved binary, LISA can measure:

- f_{GW} and $\dot{f}$ → chirp mass (without any EM data)
- Strain amplitude → combination of $\mathcal{M}_c$ and d
- Sky location → cross-match with optical surveys

Combined EM + GW observations give the full picture: component masses, radii, temperatures, orbital evolution, and distance — a powerful multi-messenger laboratory for compact binary physics.

Exercise: White Dwarf Binary Light Curves and Gravitational Waves

We will use the `ellc` package (Maxted 2016) to simulate light curves of compact double white dwarf binaries, and compute their gravitational wave properties.

The system is modeled after ZTF J1539+5027: an eclipsing double WD with $P = 6.91$ min, $m_1 = 0.61\,M_\odot$, $m_2 = 0.21\,M_\odot$, $T_1 = 50{,}000$ K, $T_2 = 10{,}000$ K, $d = 2.4$ kpc.

Tasks:

1. Use Kepler's third law to compute the semi-major axis, and express the WD radii ($R_1 = 0.013\,R_\odot$, $R_2 = 0.028\,R_\odot$) as fractions of a.
2. Use `ellc.lc()` to generate the light curve at inclination $i = 84^\circ$. Plot flux vs orbital phase. Identify the primary and secondary eclipses.
3. Compute the chirp mass, GW frequency, GW strain amplitude, and orbital period derivative $\dot{P}$.
4. Estimate the time until merger from $t_{\mathrm{merge}} \sim P/|\dot{P}|$ (order of magnitude). At what point might mass transfer begin instead?

▶ Solution

Part 2: Neutron Star Mass & Radius Measurements

We now turn from the gentle end of compact objects to the extreme: neutron stars. While white dwarfs cool passively, neutron stars reveal themselves through accretion, rotation, and gravitational waves. The common thread is that both probe the equation of state of matter under extreme conditions — electron degeneracy for WDs, nuclear matter for NSs.

The central question: **what is the mass-radius relation for neutron stars?** Unlike white dwarfs, where the M-R relation is determined by a well-understood degenerate electron EOS, the NS M-R relation depends on nuclear physics at densities several times $\rho_{\rm nuc} \approx 2.8 \times 10^{14}\,{\rm g\,cm^{-3}}$ — a regime inaccessible to terrestrial experiments. Measuring NS masses and radii is our only window into this physics.

Measurements of Neutron Star Masses and Radii

The subject is measurements of neutron star masses and radii, and what they can tell us about very dense matter. Measurement of the masses and radii of a few neutron stars is the prime goal of NASA's Neutron Star Interior Composition ExploreR (NICER) mission.

For background, you can look at:

- Miller 2013
- Miller & Lamb 2016
- Miller et al. 2019
- Miller et al. 2021

The Overall Perspective

We care about NS mass and radius measurements not just because neutron stars are awesome, but also because if we could get M and R we would learn about aspects of nuclear physics that we can't probe in laboratories. In particular, the matter in the cores of neutron stars is:

1. At several times nuclear density, and
2. Cold, in the sense that the temperature is much less than the Fermi temperature ($T_F = E_F/k$).

For such matter there should be far more neutrons than protons, which is a situation we don't encounter in the laboratory. In addition, if other particles (such as hyperons) form the bulk of the mass, their mutual interactions will be very important, and that's tough to judge in labs because we can't produce many of those exotic particles.

Above $\sim 10^{13}\,{\rm g\,cm^{-3}}$ the Fermi energy starts to contribute palpably to the total energy per neutron, and above $\sim 10^{15}\,{\rm g\,cm^{-3}}$ the total energy can exceed the rest mass energy of particles such as Λ^0, Σ^+, Δ, and Ξ^0. Interactions between these particles can change the threshold density. The central densities of realistic neutron stars range from $\sim 5 \times 10^{14}\,{\rm g\,cm^{-3}}$ to $\sim {\rm few} \times 10^{15}\,{\rm g\,cm^{-3}}$, so some of these exotic particles may indeed be energetically favorable.

The mass versus radius relation for nonrotating neutron stars depends sensitively on the equation of state: different candidate EOS predict radii from 10 to 17 km for a $2\,M_\odot$ star.

Measurements of Neutron Star Masses

The most precise and reliable measurements of neutron star masses have been made for neutron stars that are in a binary system with another star. This is because:

1. Via gravity, mass has an effect at a distance;
2. The underlying theory (Newton's laws for Keplerian motion and general relativity for post-Keplerian effects) is well-understood and tested; and
3. Many such systems are relatively clean, in the sense that there are no known complicating astrophysical effects that could potentially confuse or bias the dynamical mass measurement.

The Mass Function

If the orbital period $P_{\rm orb}$ of a binary system can be determined and the periodic changes in the line-of-sight velocity K_1 of one of the stars in the system can be measured, then to Keplerian order one can construct the **mass function** $f_1(M_1, M_2)$, which provides a lower limit on the mass M_2 of the other star. For a circular orbit:

$$M_2 \geq f_1(M_1, M_2) \equiv \frac{M_2^3 \sin^3 i}{(M_1 + M_2)^2} = \frac{K_1^3 P_{\rm orb}}{2\pi G}$$

Here M_1 is the mass of the star whose velocity is measured and i is the inclination of the orbit; $i = 0$ means that we are viewing the orbit face-on, whereas $i = 90^\circ$ means edge-on. The right-hand side is purely observable; the left-hand side contains three unknowns (M_1, M_2, i).

To uniquely determine the masses of both stars in a binary system with a known mass function, at least two additional properties of the system must be measured. The ideal systems are **double neutron star binaries**, because both objects are effectively point masses relative to their separation, and post-Keplerian effects (Shapiro delay, periastron advance, orbital decay) provide the additional constraints needed.

The Highest Measured NS Masses

The three highest neutron star masses that have been precisely measured were determined in two different ways:

- **PSR J1614-2230**: Mass determined via the Shapiro delay using radio observations. $M = 1.908 \pm 0.016\, M_\odot$ (Demorest et al. 2010).
- **PSR J0740+6620**: Also via Shapiro delay. $M = 2.08 \pm 0.07\, M_\odot$ (Cromartie et al. 2020). Currently the most massive precisely-measured NS.
- **PSR J0348+0432**: Mass determined from the white dwarf companion's spectral lines and gravitational redshift. $M = 2.01 \pm 0.04\, M_\odot$ (Antoniadis et al. 2013).

The **Shapiro delay** is a purely relativistic effect: pulses traveling through the companion's gravitational potential well are delayed relative to the flat-spacetime prediction. The delay magnitude gives the companion mass M_c; the delay variation with orbital phase gives the inclination $\sin i$.

Combined with the mass function, this yields both masses — using only clocks and geometry, with no stellar models required.

Key constraint: The existence of $\sim 2\,M_\odot$ neutron stars rules out any EOS that cannot support this mass. This immediately excludes many "soft" equations of state.

Measurements of Neutron Star Radii

Neutron stars are small and distant, so there is no prospect of angularly resolving them. We must use indirect methods.

The Blackbody Approach (and Why It Fails)

One natural possibility is to measure NS sizes the way we measure ordinary stellar radii: assume a blackbody spectrum, determine T from the spectrum, measure flux F, and distance d. Then:

$$L = 4\pi d^2 F = \sigma_{\rm SB} 4\pi R^2 T^4$$

gives us R. For ordinary stars this works well (and agrees with asteroseismological measurements). But for neutron stars, **we get absurd answers** — often less than 5 km.

The problem is that the X-ray spectral *shape* for a neutron star is close to a Planck function, but it is **not** a blackbody. The opacity is dominated by electron scattering, making the surface an inefficient emitter. To produce a given flux, the temperature must be higher than the true effective temperature. Since $R \propto T^{-2}$ in the blackbody formula, we infer a radius that is too low — often by a factor of 2 or more.

Challenges with Spectral Modeling

Even with better spectral models (atmosphere models that account for scattering), systematic uncertainties remain:

1. **Atmospheric composition**: For non-accreting NSs, is the spectrum dominated by hydrogen or helium? Helium atmospheres give ~50% larger radii than hydrogen atmospheres, with equally good fits. We often cannot distinguish between these from the spectrum alone.
2. **Non-uniform emission**: Bursts often show oscillations indicating hot spots. If non-uniform emission isn't included in the model, the total emitting area — and thus the radius — will be underestimated.

The NICER Approach: Pulse Profile Modeling

The NICER mission (Neutron Star Interior Composition Explorer, mounted on the ISS) uses a fundamentally different method. If a neutron star has hot spots on its surface, as the spot rotates, a distant observer sees an energy-dependent waveform affected by:

- **Special relativistic effects**: Doppler shifts and aberration depend on the surface velocity $v = 2\pi R\nu_{\rm spin}$, which directly probes the radius R
- **General relativistic effects**: Light bending depends on the compactness M/R. More compact stars bend light more, allowing us to see more than half the surface at once, which *reduces* the pulse amplitude

The combination of Doppler (sensitive to R) and light bending (sensitive to M/R) constrains both M and R independently.

Key advantage: Unlike spectral methods, pulse profile modeling is far more immune to systematic errors. If the model is wrong (wrong spot shape, wrong beaming pattern), the statistical fit becomes terrible — providing a clear warning. With spectral methods, you can have a good fit and still be biased.

NICER Results

- **PSR J0030+0451**: $M = 1.44^{+0.15}_{-0.14}\,M_\odot$, $R = 13.02^{+1.24}_{-1.06}\,\mathrm{km}$ (Miller et al. 2019)
- **PSR J0740+6620**: $M = 2.08 \pm 0.07\,M_\odot$, $R = 13.7^{+2.6}_{-1.5}\,\mathrm{km}$ (Miller et al. 2021)

All numbers at 1σ. The striking result: the radius is similar (~13 km) at both 1.4 and $2.1\,M_\odot$. Adding 50% more mass barely changes the size. This is consistent with a relatively stiff EOS where pressure rises sharply with density, resisting compression.

Future Methods: Gravitational Waves

The waveforms from the inspiral and merger of two neutron stars bear the imprint of tidal interactions. During the final orbits before merger, each NS raises tidal bulges on the other. How easily a NS deforms — its **tidal deformability** Λ — depends directly on the EOS: stiffer EOSs (larger radii) produce larger tidal effects that accelerate the inspiral.

The detection of GW170817 gave an upper limit to the tidal deformability, roughly implying that at 90% probability the radius of a $1.4\,M_\odot$ neutron star is less than about $13.5\,\mathrm{km}$. This complements the NICER measurements: GW tidal deformability sets an **upper** limit on radius, while NICER pulse profile modeling provides a **direct** measurement.

It may also be possible to place useful constraints on the **maximum mass** of neutron stars by combining gravitational-wave and electromagnetic observations of short gamma-ray bursts. Applying this logic to GW170817, Margalit & Metzger found a maximum mass for a nonrotating neutron star of $\sim 2.17\,M_\odot$, with similar limits of $M_{\max} \sim 2.2\text{–}2.3\,M_\odot$ from numerical models.

The emerging picture from all methods combined:

- **Lower limit on $M_{\max}$**: $\gtrsim 2.08\,M_\odot$ (from PSR J0740+6620)
- **Upper limit on $M_{\max}$**: $\lesssim 2.3\,M_\odot$ (from GW170817 + kilonova modeling)
- **Radius at $1.4\,M_\odot$**: $\sim 11\text{–}14\,\mathrm{km}$ (from NICER + GW170817)

With more NS merger events from LIGO/Virgo/KAGRA and continued NICER observations, these constraints will continue to tighten.

Exercise 1: The Mass Function

The mass function for a circular binary is:

$$f_1(M_1, M_2) = \frac{M_2^3 \sin^3 i}{(M_1 + M_2)^2} = \frac{K_1^3 P_{\mathrm{orb}}}{2\pi G}$$

Consider a neutron star X-ray binary with $P_{\mathrm{orb}} = 4.8\,\mathrm{hr}$ and $K_1 = 300\,\mathrm{km/s}$ (measured from the companion star).

Task:

1. Compute the mass function f_1.
2. Plot M_{NS} (the neutron star mass) as a function of inclination i (from 20° to 90°) assuming a companion mass of $M_1 = 0.5\,M_\odot$. Solve the cubic equation numerically.
3. For what range of inclinations is the compact object consistent with being a neutron star ($M < 3\,M_\odot$)?

▶ Solution

In-Class Assignment: X-ray Burst Timing

The Rossi X-ray Timing Explorer (RXTE) has very high time resolution and has been used to observe a large number of X-ray sources. We'll look at the data from the low-mass X-ray binary 4U 1728-34—this is an X-ray burst system.

In Strohmayer et al. 1996 it was shown that the neutron star spin rate can be seen in the Fourier transform of the lightcurve of the burst. Here we repeat the analysis.

We thank Tod Strohmayer for sharing the data from that paper.

Learning Objectives

- Methods to reduce and visualize XRB lightcurve data
- Application of FFT to astronomical data
- Deriving NS spin rates from XRB data

Download the following data locally: 4u1728_burstdata.txt.

a. - Flatten the data, prepare for analysis

1. Load the data using numpy `loadtxt`
2. Flatten the data (`.flatten()`)
3. Obtain the total length of the 1-d array and define it as `N` for later
4. Plot the data using matplotlib.

> What are the x-axis and y-axis?

b. - Bin the data, replot

Similar to the paper, lets bin the data using the `.reshape()` operator on the original flattened data and floor division (`//`) for a choice of bin size = 256 to create a new array `binned_data`.

```
{note}
Reshape syntax is numpy.reshape(rows,columns)
```

Steps will be

1. Create a new array that is equal to the original data reshaped to be (N // 256, 256) and the bins summed (.sum) along the 1 axis (rows).
2. Plot the new binned lightcurve data

> Does this plot match Figure 1 in the paper?

c. -Fast-Fourier Transform of the Data

Using the raw flattened data (non-binned),

1. compute a one-dimensional discrete Fourier Transform using numpy to obtain the Fourier coefficients C_k.

2. Next, compute the physical frequencies using numpy.fft.rfftfreq. You will need to pass the total number of samples from earlier `N` then multiply by N/T where $T = 32$ s and is the total duration of the signal.

3. Using C_k and `kreq` plot the power spectrum as a function of frequency: (`kfreq` , `np.abs(c_k)**2 * 2 / N`) using log-log.

> Can we identify any excess power in the spectrum?

d. - Bin the FFT data to improve Signal to noise

The original paper binned the FFT by 8, lets do the same here.

1. Bin the data using the provided lines.
2. Replot the data similar as in c. Using a log-log plot.

> Now, can we identify any excess power in the spectrum?

e. - Plot the final frequency

1. On a linear-linear plot, zoom into the suspected frequency identified.

> Does this match the result from the paper? What does this frequency correspond to?

In-Class Assignment: Neutron Star Mass-Radius Relation

Credit: Michael Zingale.

Learning Objectives

- Develop a qualitative understanding of the M-R relationship for neutron stars
- Identify the impact of the compressibility modulus K on the M-R relation and the maximum mass

Here's a general integration class that implemented 4th order Runge-Kutta. It takes a `rhs` function of the form:

```
rhs(r, Y, rho_func)
```

that returns the derivatives $d\mathbf{Y}/dr$ for each of the variables in the array $\mathbf{Y}$. Here, `rho_func` is a function `rho_func(pres)` that takes the pressure and returns the density according to our EOS.

Some physical constants (CGS)

Here's the righthand side function for the TOV equation

Here's the righthand side function for Newtonian HSE

Degenerate neutrons

For zero-T neutron degeneracy, the number density of neutrons is just the integral of the distribution function, $n(p)$, over all momenta:

$$n = \int n(p) d^3p = 4\pi \int_0^\infty n(p) p^2 dp$$

but at zero temperature the distribution function is just a step function:

$$n(p) = \frac{2}{h^3} \begin{cases} 1 & \text{if } p < p_F \\ 0 & \text{otherwise} \end{cases}$$

This gives:

$$n = \frac{\rho}{m_u} = \frac{8\pi}{3h^3} p_F^3 = \frac{8\pi}{3} \left(\frac{m_u c}{h}\right)^3 x_F^3$$

where $x_F = p_F/(m_u c)$.

We can then compute the pressure as:

$$P = \frac{1}{3} \int n(p) v p d^3p$$

where we now need to use the general expression for velocity,

$$v = \frac{p}{m_u} \left[1 + \left(\frac{p}{m_u c}\right)^2\right]^{-1/2}$$

This gives:

$$P = \frac{\pi}{3} \left(\frac{m_u c}{h}\right)^3 m_u c^2 f(x)$$

with

$$f(x) = x(2x^2 - 3)(1 + x^2)^{1/2} + 3 \sinh^{-1}(x)$$

A polytropic nuclear EOS

A simple polytopic approximation to a nuclear equation of state (the SLy4 EOS) is:

$$P = K\left(\frac{\rho}{\rho_0}\right)^\gamma$$

If we choose $K = 3.5\text{MeV fm}^{-3}$ and $\rho_0 = 150\text{MeV fm}^{-3}$, we approximate the behavior of the SLy4 nuclear EOS (thanks to Tianqi Zhao for these approximations) with $\gamma = 2.5$.

In CGS units, using $\gamma = 2.5$, we can take $K = 4.80 \times 10^{-3}$ erg cm^{-3} and write this as:

$$P = K\rho^\gamma$$

Notebook begins here

a. - The "Chandrasekhar mass" of a Neutron Star

The Chandrasekhar mass, the maximum mass that electron degenerate objects (of fixed μ_e) can have before becoming gravitationally unstable and collapsing, is given by HKT 3.67:

$$M_{\text{Ch}} = 1.44\left(\frac{2}{\mu}\right)^2 M_\odot$$

For electrons, the μ was μ_e, as in white dwarfs. But, let's use this to compute the maximum mass of a neutron star, where now $\mu = 1$ for neutrons.

1. Compute the maximum mass for a NS using the equation above.

b. - Case I: Newtonian gravity, neutron degeneracy

Now, let's look at the M-R relation for a Neutron star where we will use our usual HSE, but with pressure derived from n degeneracy pressure.

1. Evaluate the cell below for a M-R relation using our usual non-GR HSE, but with n degeneracy pressure.
2. Plot the resulting M-R and label the maximum allowed NS mass from this result.

> Compare with the result of a. Do they match?

We'll loop over the central density and integrate the neutron star structure and store the mass and radius corresponding to each central density.

We see that we are approaching that maximum mass as $R \to 0$, as expected.

c. - Case II: GR gravity (TOV) + neutron degeneracy

1. Evalue the cell below, plot the resulting R vs. M plot.
2. Compute the maximum allowed mass under these assumptions.

> Is this mass less or more than the previous? Qualitatively, what lead to the change in the maximum mass from looking at the R-M profile?

Now, adding GR, we see that there is a maximum mass for the neutron star, but it is quite low.

d - Case III: GR gravity (TOV) + nuclear EOS

Next, let's use a more accuracte equation of state using an analtical form for the nuclear EoS defined above.

1. Evaluate the cell below, plot the result.
2. Compute the maximum NS mass allowed under these assumptions.

> Qualitatively describe the new M-R profile when changing the EoS compared to the previous profile in c.

In this final case, we get a much more reasonable maximum neutron star mass. We also see that for some values of radius there are multiple masses (the M-R relation is double-valued) -- this is okay. However, at the very highest central densities, we get a smaller mass than the maximum at a smaller radius -- that is not physical, and is dynamically unstable.

e - The impact of the compressibility modulus

1. Plot the results of `c` and `d` on the same plot.
2. Looking only at the respective maximum mass for each profile, label which relation suggests a "softer" EoS and which suggests a "stiffer" EoS.

> Qualitatively describe the difference between the "softer" and "stiffer" EoS, what that means for the incompressibility of the matter and the maximum mass.